

From Language to Action: Can LLM-Based Agents Be Used for Embodied Robot Cognition?

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Introduction

Motivation

- Robots in everyday environments need **task planning, execution monitoring, and failure recovery**
- Traditional cognitive architectures [1, 2] require extensive engineered knowledge and many dedicated components
- LLMs exhibit emergent capabilities in reasoning, language understanding, and task execution [3, 4]
- Prior LLM-robot work [5, 6] relies on symbolic world representations or code generation; we investigate zero-shot tool-calling

Research Question

Can an LLM (used *zero-shot*) serve as the cognitive core of an embodied robot in a 3D household environment?

Contributions

- Cognitive architecture** embedding an agentic LLM with episodic memory, working memory, and a tool-calling interface for navigation and manipulation
- Empirical evaluation** on two household tasks across four LLMs (GPT-4.1, Claude 4 Sonnet, Qwen3 Coder 480B, DeepSeek V3.1)

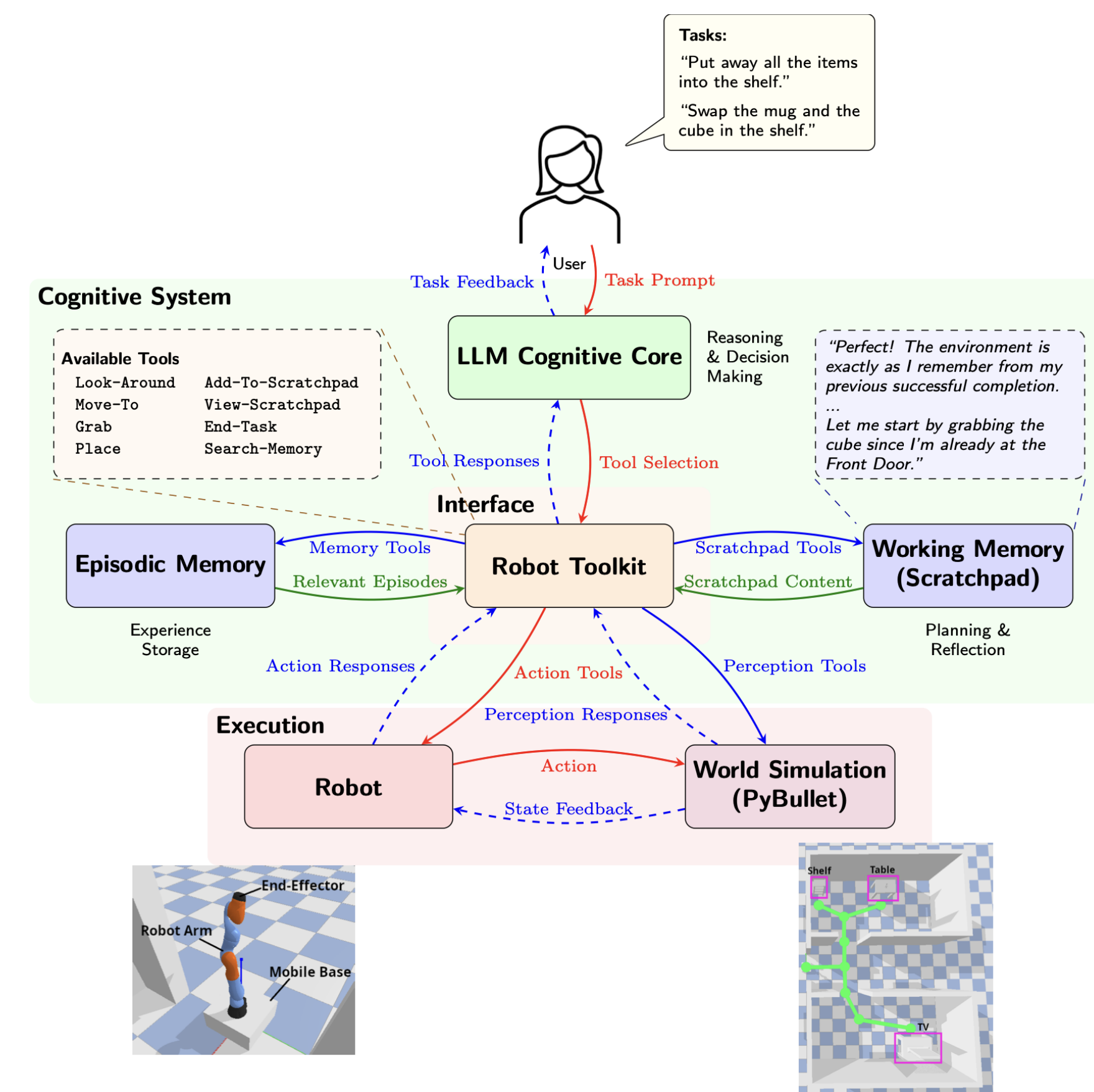


Fig. 1: System overview

System Architecture

The robot is embodied as a **mobile manipulator** (omnidirectional base + 7-DOF arm) operating in a PyBullet-simulated household environment with a kitchen, living room, and hallway.

Memory Architecture

- Working memory:** LLM context + scratchpad for immediate task reasoning and planning
- Episodic memory:** ChromaDB vector store of past execution summaries, semantically searchable for relevant prior experiences

Tools (callable actions for the LLM)

- Look-Around — perceive nearby objects and current location
- Move-To — navigate to a semantic map location via A*
- Grab — attach object to end-effector (if within reach)
- Place — release held object at a target location
- Add-To-Scratchpad / View-Scratchpad — plan and reason
- Search-Memory — retrieve relevant past experiences
- End-Task — report task outcome; result saved to episodic memory

Chain-of-thought prompting elicits reasoning; replanning emerges naturally from LLM inference without explicit enforcement.

References

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- [5] L. Manling et al. Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making. In *38th Conf. Neural Information Processing Systems (NeurIPS)*, pages 100428–100444, 2024.
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Evaluation

Tasks (static setup, 3D simulated household)

- T1 – Placing items:** place a mug, box, and cube into a three-shelf cupboard
- T2 – Swapping items:** swap mug and cube (requires reasoning about a temporary storage location; executed after T1)

Execution Success Rate

	GPT-4.1	Claude 4	Qwen3 480B	DeepSeek V3.1
T1	100%	100%	80%	100%
T2	44.4%	100%	66.2%	75.5%

All models reliably complete T1; T2 is more challenging (requires multi-step reasoning about a temporary swap location).

Hallucinated Success

All models *overestimate* their success — they believe they have succeeded even when the world state is incorrect.

Effect of Episodic Memory

- Memory *reduces tool calls* on average for all models and tasks
- Success rate improvement is modest and model-dependent; hallucinated memory labels limit benefits

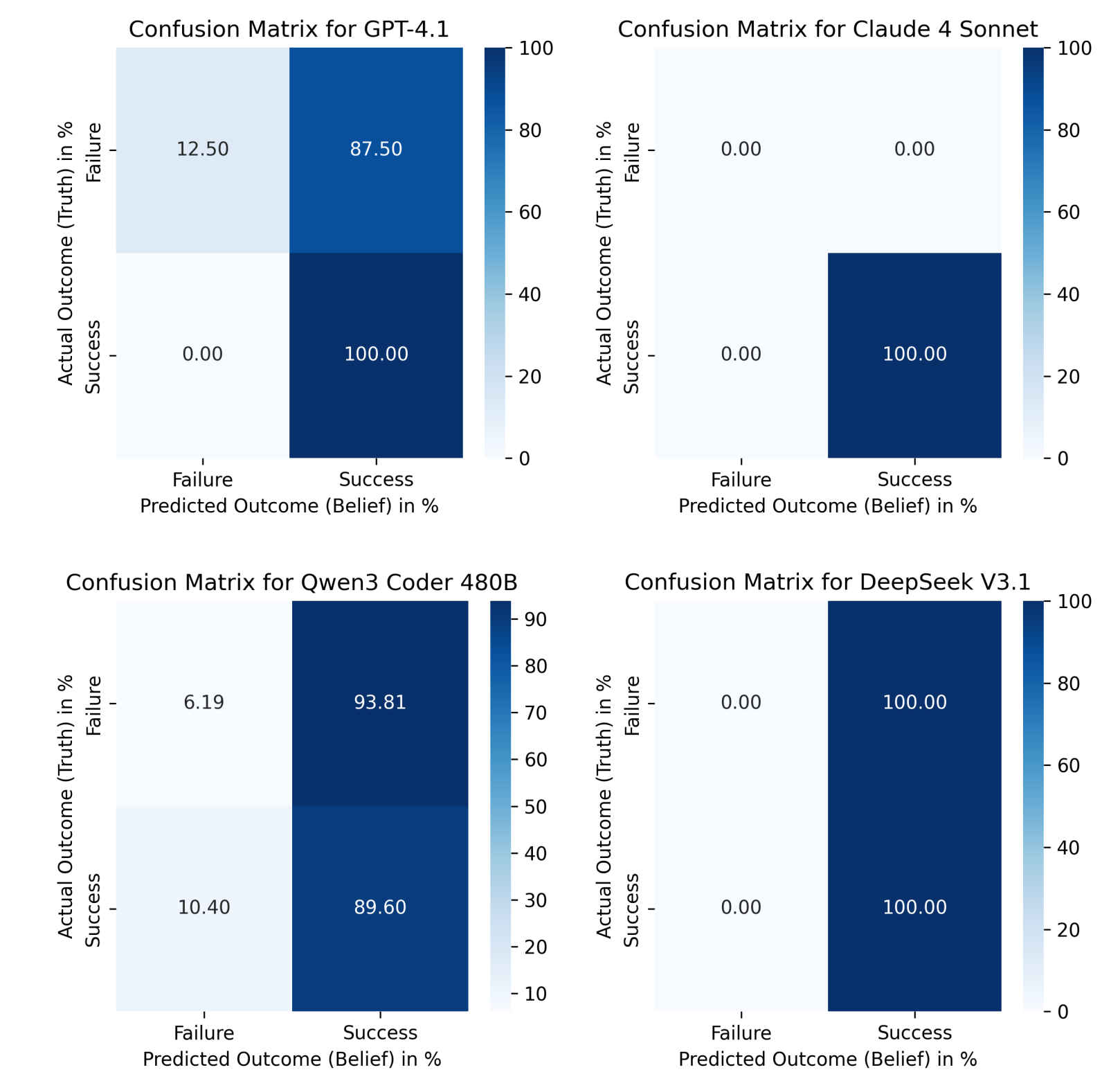


Fig. 3: Believed vs. actual success for all four models

Discussion & Conclusions

Key Findings

- LLMs can reason about spatial layouts, sequence tool calls, and **recover from failures without explicit replanning logic**
- Episodic memory provides modest improvements; benefits are limited by **hallucinated success labels** corrupting the memory store
- Models misinterpret ambiguous instructions (e.g., attempting to grab a TV when told to “put all objects away”)
- Smaller models (~24B parameters) cannot coherently chain actions; large frontier models are necessary

Limitations

- Overconfidence:** agents incorrectly believe they succeeded, impairing episodic memory quality
- Task refusal:** after completing one task, models sometimes hallucinate that a subsequent task is already done
- High latency** of cloud-hosted models (30–150s/request) limits real-robot reactivity
- Evaluation is simulation-only; real-robot deployment not yet validated

Future Work

- Couple episodic memory writes with an **external fact-checker** (human-in-the-loop or dedicated model) to eliminate hallucinated labels
- Automated evaluation pipeline for broader task/model/platform benchmarking
- Richer simulated environments (dynamic obstacles, multi-agent) and **physical robot deployment**
- Systematic comparison with traditional cognitive architectures (SOAR, ACT-R, KnowRob)

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